

Player Tracking in Sports: Multi-View Tracking and 3D Reconstruction

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1 Introduction

Vision-based tracking provides a scalable alternative to expensive sports hardware, yet basketball’s rapid movements, mutual occlusions, and floor specularities pose severe challenges. While modern pipelines have shifted from classical background subtraction [3] toward tracking-by-detection frameworks [1], maintaining identity consistency during close-range crossings remains a bottleneck. This work introduces an end-to-end multi-view pipeline that bridges these gaps by deploying a domain-adapted YOLOv26 detector [5] coupled with an appearance-precedent DeepSORT tracker [6, 8]. By leveraging multi-camera geometry and SVD-based triangulation, the system successfully resolves 2D ambiguities to reconstruct smooth, continuous 3D trajectories across the court.

2 Detection

The evolution of object detection paradigms in our framework moves from classical computer vision to domain-adapted deep learning. As an initial baseline, a Mixture of Gaussians (MOG2) background subtraction approach [3] enhanced with CLAHE preprocessing [9] proves highly vulnerable to court floor specularities, player stasis, and dynamic audience backgrounds. Transitioning to an object detection paradigm via a COCO-pre-trained YOLOv11 model [5] substantially mitigates these spatial errors by extracting robust semantic features. However, significant domain gaps persist with the off-the-shelf model: the basketball is entirely absent from the pre-trained taxonomy, and players and referees are uniformly grouped under a single “person” class, precluding downstream identity distinction and necessitating a fine-tuned, domain-specific architecture.

2.1 Fine-Tuned YOLOv26 with Multi-Pass Inference

To bridge these domain gaps, we adopt the YOLOv26 architecture and fine-tune its backbone on a domain-specific annotated dataset. The model is trained to recognise each player individually, as well as the ball, so identity is resolved directly by the detector.

However, because the ball occupies a minimal pixel area, standard single-pass inference downsamples it too aggressively for reliable localization. We circumvent

this spatial resolution trade-off via a dual-pass inference strategy: a primary pass optimized for player detection, and a parallel high-resolution pass configured with a permissive confidence threshold specifically tailored to isolate the ball. The resulting detections are subsequently unified into a single composite output.

Finally, to eliminate duplicate bounding boxes when a player is detected under multiple conflicting identities, we substitute standard non-maximum suppression with a class-agnostic variant [4]. This suppresses redundant detections by spatial overlap alone, ensuring a single clean bounding box per player is passed to the downstream tracker.



Fig. 1. Sample detection output from fine-tuned YOLOv26 with class-agnostic NMS on cam_13 (three selected frames).

3 Tracking

To maintain continuous trajectories across frames, the temporal tracking framework transitions from a purely spatial baseline to an appearance-robust architecture. As a foundational baseline, we employ Simple Online and Realtime Tracking (SORT) [1], which propagates tracks via a Kalman filter and couples them to incoming detections using the Hungarian algorithm based on Intersection-over-Union (IoU) proximity. However, SORT’s reliance on spatial overlap causes frequent identity switches when players cross paths. To overcome this, we transition to an appearance-precedent DeepSORT framework with robust tracking and label arbitration mechanisms to maintain identity consistency.

3.1 DeepSORT

To minimize identity swaps, our final tracking framework utilizes DeepSORT [6], which augments the traditional Kalman filter motion model with appearance cues. For each detection, a lightweight OSNet architecture [8] extracts an appearance embedding, appended to a rolling gallery for each active trajectory. Gating and association proceed via a matching cascade that prioritizes appearance similarity over spatial overlap.

On top of this base we introduce several improvements. The **confidence-aware Kalman update (NSA)** reduces state-correction jitter: instead of weighing every bounding-box update uniformly, we scale the measurement covariance by the detection confidence. **Low-confidence recovery (ByteTrack)**

[7] matches unmatched tracks against low-confidence detections using spatial overlap, maintaining continuity through partial occlusions. To prevent template corruption, these noisy boxes are excluded from appearance-embedding updates. **Track resurrection** mitigates prolonged occlusions by retaining the appearance galleries of recently terminated tracks for a brief window. If a newly detected player closely matches an inactive gallery, their original identity is restored rather than initializing a new track. Finally, **coasting** bridges brief gaps by reporting a track’s Kalman-predicted position while unmatched, provided the prediction remains within the frame bounds and does not overlap an already assigned bounding box.

A final **label resolution** step unifies the detector’s per-frame class predictions across each trajectory. Each track is assigned its confidence-weighted majority label, with identity conflicts between tracks resolved in favor of the higher-confidence prediction. The ball category is exempt from this unique-identity arbitration.



Fig. 2. Sample tracking output from DeepSORT with resolved identities on cam_13 (three selected frames).

4 3D Reconstruction

Having recovered the per-camera 2D trajectories, we lift them into a unified 3D scene representation shared by all three cameras. This geometric lifting relies on intrinsic matrices known *a priori*, while extrinsic parameters and lens distortion are estimated by solving the Perspective-n-Point (PnP) problem, mapping manually annotated 2D court landmarks from the initial frame to their known 3D world coordinates.

For triangulation, detections are grouped by identity and frame across cameras and undistorted. The 3D reconstruction then bifurcates based on the distinct physical constraints of the targets. Because players move on the court floor, each camera’s bottom bounding-box pixel is back-projected into a ray whose intersection with the floor plane yields a ground position. This floor constraint enables localization from a single view during occlusions, while multi-view estimates are averaged after outlier rejection. Conversely, the airborne ball is triangulated from the rays of at least two cameras using the Direct Linear Transform (DLT) solved

via Singular Value Decomposition (SVD). Views with high reprojection errors are iteratively discarded and the point re-triangulated.

Finally, trajectory smoothing removes residual noise from detection jitter and calibration errors. Each 3D trajectory is filtered forward via a Kalman filter and refined by a Rauch–Tung–Striebel (RTS) backward pass. This bidirectional processing uses past and future observations to correct states, reject spatial jumps, and split trajectories across extended gaps to prevent false extrapolation.

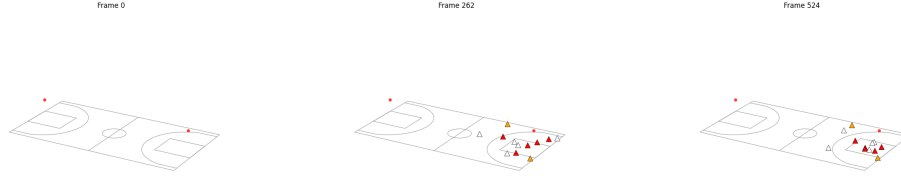


Fig. 3. Reconstructed 3D player and ball positions projected onto the court plane after RTS smoothing.

5 Evaluation Metrics

We evaluate each stage of the pipeline separately. **Detection** is measured with precision, recall, and the F_1 -score, with true positives defined at an Intersection over Union (IoU) threshold of 0.5.

Tracking is assessed with IDF1 and HOTA [2]: IDF1 captures identity consistency, while our primary metric, HOTA, is the geometric mean of detection accuracy (DetA) and association accuracy (AssA), averaged over a range of localization thresholds. We favour HOTA over traditional metrics such as MOTA, which are biased toward detection errors and poorly reflect tracking continuity.

3D geometry accuracy is evaluated using two primary metrics. The **reprojection error** measures the pixel distance between a triangulated 3D point and its original 2D detection across observing cameras, indicating geometric consistency. The **trajectory error** quantifies tracking accuracy by comparing these 3D trajectories against ground-truth 2D data across all shared frames, reported via the Average (ADE), Final (FDE), and Median (MTE) displacement errors, alongside the number of trajectory fragments.

6 Results

We summarise here the final tracking and trajectory results; the full set of detection, tracking, and geometry tables, together with the per-stage qualitative

outputs and all pipeline parameters, is reported in the appendix (Appendices B–D).

Table 1 reports the metrics of our final tracker, DeepSORT, across the three cameras. DeepSORT maintains identities reasonably well across all views, with the best scores on cam_4, and the similar DetA and AssA values indicate that detection recall, rather than association, is the main factor limiting tracking quality on this short clip.

Table 1. Final tracking results (DeepSORT).

Camera	IDF1	HOTA	DetA	AssA
cam_13	0.569	0.434	0.469	0.405
cam_2	0.610	0.480	0.453	0.511
cam_4	0.638	0.521	0.553	0.495

The reconstructed 3D tracks are then projected back onto the annotated 2D trajectories, with the resulting errors reported in Table 2. These errors are large across all views, although the low fragment count shows that the tracks still cover the annotated span almost without gaps. The dominant source of error is calibration accuracy: a bias in the estimated camera pose propagates directly into a constant positional offset in the reconstructed trajectories.

Table 2. Trajectory error of projected 3D tracks vs. ground-truth 2D annotations.

Camera	ADE (px)	FDE (px)	MTE (px)	Fragments
cam_13	976.8	849.8	1047.5	2
cam_2	1066.0	1279.1	1181.4	5
cam_4	1838.2	1689.1	1899.9	5

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A Dataset

The pipeline is evaluated on a short basketball sequence recorded at the Sanbapolis facility by three synchronised cameras (cam_13, cam_2, and cam_4). The clip spans 525 frames at 25 fps (about 21 seconds) and covers a full FIBA court of 28×15 m.

Each camera’s intrinsic parameters are provided. The extrinsics are estimated by solving the Perspective-n-Point problem from manually selected court landmarks on the first frame, with lens distortion modelled directly inside the solver.

For evaluation, ground-truth 2D bounding boxes are annotated on every fifth frame. The detector recognises the ball as a single class and each player as its own identity class; it was fine-tuned on a separate dataset from the same domain, covering different actions of the same game.

B Pipeline Parameters

The following tables list the parameters used at each stage of the pipeline.

Table 3. MOG2 background-subtraction parameters.

Parameter	Value
Learning rate	−1 (adaptive)
History (frames)	1000
Variance threshold	25
Opening kernel (px)	7
Closing kernel (px)	11
Min. contour area (px ²)	1500
Max. contour area (px ²)	200000

Table 4. YOLO detection parameters (pre-trained, fine-tuned multi-pass, and NMS).

Parameter	Value
<i>Pre-trained YOLOv11</i>	
Confidence threshold	0.4
Built-in NMS IoU	0.45
Inference size (px)	640
<i>Fine-tuned YOLOv26 (multi-pass)</i>	
Player confidence	0.3
Player low-conf band	0.15
Ball confidence	0.2
Player inference size (px)	1280
Ball inference size (px)	1600
<i>Class-agnostic NMS</i>	
IoU suppression threshold	0.75

Table 5. Tracking parameters (SORT and DeepSORT).

Parameter	Value
<i>SORT</i>	
Max. IoU distance	0.8
Max. age (frames)	60
Confirmation hits	2
<i>DeepSORT</i>	
Max. IoU distance	0.8
Max. appearance distance	0.2
Max. age (frames)	60
Confirmation hits	2
Feature budget per track	100

Table 6. 2D trajectory smoothing, triangulation, and 3D smoothing parameters.

Parameter	Value
<i>2D trajectory smoothing</i>	
Max. interpolated gap (frames)	5
EMA weight α	0.5
<i>Triangulation</i>	
Ground inlier gate (mm)	600
Ball reprojection gate (px)	30
Minimum views	2
<i>3D trajectory smoothing</i>	
Position noise std (mm)	30
Velocity noise std (mm/frame)	60
Measurement noise std (mm)	80

C Detailed Results

Table 7. Detection results across cameras (IoU threshold = 0.5).

Method	Camera	Precision	Recall	F_1	Mean IoU
MOG2	cam_13	0.471	0.262	0.337	0.651
	cam_2	0.345	0.408	0.374	0.663
	cam_4	0.328	0.293	0.309	0.700
YOLO pre-trained	cam_13	0.834	0.602	0.699	0.748
	cam_2	0.819	0.781	0.800	0.785
	cam_4	0.927	0.807	0.862	0.808
YOLO fine-tuned	cam_13	0.872	0.732	0.796	0.758
	cam_2	0.835	0.761	0.796	0.757
	cam_4	0.923	0.822	0.870	0.796

Table 8. Identity tracking metrics (IoU threshold = 0.5).

Tracker	Camera	IDP	IDR	IDF1
SORT	cam_13	0.660	0.514	0.590
	cam_2	0.743	0.607	0.668
	cam_4	0.768	0.605	0.679
DeepSORT	cam_13	0.588	0.533	0.569
	cam_2	0.617	0.603	0.610
	cam_4	0.670	0.605	0.638

Table 9. HOTA tracking metrics.

Tracker	Camera	HOTA	DetA	AssA	LocA
SORT	cam_13	0.438	0.484	0.398	0.797
	cam_2	0.517	0.515	0.522	0.794
	cam_4	0.542	0.575	0.514	0.829
DeepSORT	cam_13	0.434	0.469	0.405	0.769
	cam_2	0.480	0.453	0.511	0.761
	cam_4	0.521	0.553	0.495	0.798

Table 10. Reprojection error of triangulated 3D points.

Camera	Mean (px)	Median (px)	RMSE (px)	Acc@5px	Acc@10px	Acc@20px
cam_13	51.3	28.9	87.7	0.135	0.221	0.349
cam_2	140.0	94.5	278.7	0.028	0.047	0.094
cam_4	135.0	50.0	296.2	0.168	0.214	0.295

Table 11. Trajectory error of projected 3D tracks vs. ground-truth 2D annotations.

Camera	ADE (px)	FDE (px)	MTE (px)	Fragments
cam_13	976.8	849.8	1047.5	2
cam_2	1066.0	1279.1	1181.4	5
cam_4	1838.2	1689.1	1899.9	5

D Qualitative Outputs

The figures below are the per-stage visual outputs produced by the pipeline notebook on cam_13.

**Fig. 4.** Raw input frame.



Fig. 5. MOG2 foreground mask.



Fig. 6. MOG2 detections overlaid on the frame.



Fig. 7. Pre-trained YOLOv11 detections.



Fig. 8. Fine-tuned YOLO merged player and ball detections.



Fig. 9. Detections after class-agnostic NMS.



Fig. 10. SORT tracks.



Fig. 11. SORT tracks after label resolution.



Fig. 12. DeepSORT tracks.



Fig. 13. DeepSORT tracks after smoothing and label resolution.

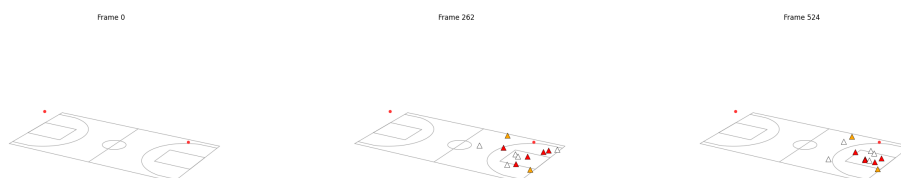


Fig. 14. Triangulated 3D reconstruction (top view).

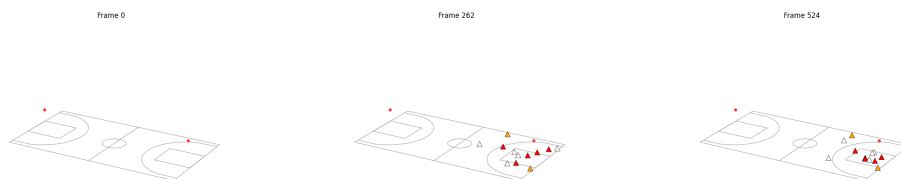


Fig. 15. Smoothed 3D reconstruction (top view).